

Xiao Tan

B.S. Student in Computer Science, University of North Carolina at Chapel Hill
Undergraduate Research Assistant, Hardware Security Lab @ UNC
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RESEARCH INTERESTS

Hardware Security • Formal Methods • Computer Architecture

EDUCATION

University of North Carolina at Chapel Hill , Chapel Hill, NC, United States	August 2023 — May 2027
Bachelor of Science in Computer Science	Cumulative GPA: 3.94/4.00
Dean's List (2023 Fall - 2026 Spring)	

PUBLICATIONS

CHARGE: Leveraging CWE Hierarchies for Hardware Security SystemVerilog Assertion Generation.

Xiao Tan, Cynthia Sturton.

Accepted to the IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2026.

Hardware Security Benchmarks for Open-Source SystemVerilog Designs.

Jayden Rogers, Niyaz Shakeel, **Xiao Tan**, Samantha Espinosa, Divya Mankani, Cade Chabra, Kaki Ryan, Cynthia Sturton.
In *Proceedings of the IEEE Secure Development Conference (SecDev)*, 2025.

RESEARCH EXPERIENCE

Hardware Security Lab @ UNC	Chapel Hill, NC, United States
PI: Professor Cynthia Sturton	Jan. 2025 – Present

CHARGE (ICCAD 2026)	Fall 2025 – Present
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- Designed and implemented CHARGE, an automated framework that generates hardware security SystemVerilog Assertions (SVAs) directly from RTL using CWE hierarchies and large language models, eliminating the need for trusted specifications.
- Designed a three-stage pipeline comprising asset identification, behavior mining, and SVA generation. Leveraged the hierarchical structure of the CWE taxonomy to improve security-critical asset identification.
- Evaluated CHARGE on the Hack@DAC18, 19, and 21 benchmark suites, detecting 27 of 42 known hardware security bugs and identifying one previously unknown bug.
- Currently extending CHARGE to reduce false positives and improve the quality of generated security assertions.

Hardware Security Benchmarks (SecDev 2025)	Spring 2025
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- Developed SystemVerilog Assertions (SVAs) to verify security properties of open-source System-on-Chip (SoC) designs using Cadence JasperGold Formal Property Verification (FPV).
- Analyzed manually written security properties using the CWE taxonomy to assess coverage across hardware security weakness classes.

Research Infrastructure	Fall 2025
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- Collaborated on maintaining SylQ-SV, a symbolic execution engine for SystemVerilog, by resolving parser compatibility issues between PyVerilog and PySlang.

ACADEMIC PROJECTS

Verification of Access Control Enforcement in AXI Interconnect	UNC Chapel Hill
<i>COMP 637: Formal Methods for System Security</i>	Spring 2026

- Verified access-control enforcement in an AXI-Lite interconnect extracted from the Hack@DAC21 OpenPiton SoC using assertion-based verification.
- Translated 21 access-control property templates from the AKER framework (Restuccia et al., ICCAD 2021).
- Applied Cadence JasperGold FPV and SPV engines to perform model checking and information-flow tracking across controller-peripheral interactions.
- Successfully instantiated 17 properties, with 12 satisfied and covered; analyzed 5 violations and identified two inaccuracies in the original AKER templates.

FPGA Rhythm Game on Custom MIPS Processor*COMP 541: Digital Logic and Computer Design*

UNC Chapel Hill

Spring 2026

- Designed and implemented a custom single-cycle MIPS processor in SystemVerilog and deployed it on a Nexys A7 FPGA. Integrated VGA graphics, accelerometer, keyboard, and audio output through memory-mapped I/O (MMIO).
- Developed a real-time rhythm game inspired by Trombone Champ, featuring VGA-rendered scrolling notes, accelerometer-based pitch control, keyboard-triggered note timing, score/combo tracking, and discrete audio feedback.
- Implemented the game in C and manually translated the program into MIPS assembly for execution on the custom processor.
- Built the complete hardware/software stack, including peripheral drivers, graphics rendering, input handling, and audio generation, demonstrating hardware/software co-design on an FPGA platform.

Network Traffic Application Classification via Sequence Models*COMP 431: Internet Services and Protocols*

UNC Chapel Hill

Fall 2025

- Collected real-world network traffic traces using `tcpdump` and constructed a dataset across multiple application types (web, video, gaming, VoIP)
- Extracted structural traffic features including packet sizes, inter-arrival times (IAT), and MTU-based metrics
- Evaluated clustering methods (K-Means, DBSCAN) and observed limited separation across application classes
- Applied sequence models (BiLSTM/LSTM) to capture temporal patterns in traffic flows, achieving ~96% classification accuracy

PRESENTATIONS

Poster: Leveraging CWE Hierarchies for Hardware Security SystemVerilog Assertion Generation*CRA UR2PhD Undergraduate Mentoring Workshop and Research Showcase*

April 2026

Selected participant (travel fully funded).

SELECTED COURSES

Formal Verification for System Security; Digital Logic and Computer Design; Computer Security Concepts; Models of Languages and Computation; Operating Systems (Fall 2026); Compilers (Fall 2026)

AWARDS

Summer Undergraduate Research Fellowship (SURF)

UNC Chapel Hill

Awarded a \$4,000 summer research stipend to work on automated hardware security assertion generation.

March 2026

TEACHING EXPERIENCE

UNC-Chapel Hill Department of Computer Science*Undergraduate Teaching Assistant*

Chapel Hill, NC, United States

June 2025 — December 2025

- Supported 300+ students in COMP 301 (Object-Oriented Programming and Design Patterns) through 10+ weekly office hours across Summer and Fall 2025
- Assisted students in debugging complex Java codebases involving OOP, MVC architecture, and concurrency
- Designed programming assignments and exam review resources as part of the Additional Materials Committee

OTHER EXPERIENCE

UNC-Chapel Hill ITS Service Desk*Student Assistant - Walk-in Services*

Chapel Hill, NC, United States

June 2025 — December 2025

- Diagnosed and resolved 50+ monthly technical support tickets across hardware, software, and network issues
- Achieved a high first-contact resolution rate by delivering clear, step-by-step technical solutions
- Documented recurring issues to streamline troubleshooting workflows and improve support efficiency

SKILLS

- **Hardware Description Languages:** Verilog, SystemVerilog
- **Formal Verification:** SystemVerilog Assertions (SVA), Cadence JasperGold FPV/SPV
- **Programming Languages:** Verilog, SystemVerilog
- **Tools:** Vivado, Git, Linux